Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18IS54]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

COMPILER DESIGN

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. With a neat diagram, explain the different phases of compiler. **07**

b. How are symbol tables managed? Who are the cousins of compiler? How compiler will manage the data? **07**

2. a. What is lexical analyzer? Bring out the issues in lexical analyzer. Explain with an example. **07**

b. What are the different steps of Boot Strapping? With an example, explain the Buffer Pairs and Sentinels of an expression. **07**

3. a. i. With a clear example, explain the Top Down Parsing. **08**

ii. Differentiate Top Down Parsing and Bottom up Parsing

b. What is shift reduce parsing? How it is different from Operator precedence parsing perform Shift reduce parsing for id * id using stack operations. **06**

OR

4. a. i. Construct a predictive parse table for the expression id + id * id. **08**

ii. Give all the moves made by predictive parser for id + id * id using First and Follow

b. Construct operator precedence table and accept input string " **id+id*id**" and also show the stack operations **06**

5. a. What is YACC? Explain how to use it in constructing the parser. **05**

b. E -> E + T | T **09**

T -> T * F | F

F-> (E) | id

Construct the LALR (1) Parsing table?

OR

6. a. Explain the 3 types of Implementation of Three Address codes with clear example **07**

b. What are storage allocation strategies? Explain different types of strategies with example. **07**

7.	**a.**	What are dominators? How existence of dominators tree can be used in compiler design? Explain with example.	**08**
	b.	With a neat illustration, Explain the Natural loop algorithm of a back edge.	**06**

OR

8.	**a.**	What are target machines? With different instructions, explain how instruction sequences can be validated.	**06**
	b.	Explain Peephole optimization with different passes, jumps and how it can be handled.	**08**